Hierarchical Classification of Moving Vehicles Based on Empirical Mode Decomposition of Micro-Doppler Signatures

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***Abstract*—A novel method is proposed for moving wheeled vehicle and tracked vehicle classification using micro-Doppler features from returned radar signals within short dwell time. In this method, an adaptive analysis technique called Empirical Mode Decomposition (EMD) is utilized to decompose the motion components of moving vehicles, and a hierarchical classification structure using the decomposition results of returned signals is proposed to discriminate the two kinds of vehicles. The first stage of the structure elementarily identifies the tracked vehicle data by checking the existence of its unique feature and a further classification via our proposed features based on EMD is implemented in the second stage by using Support Vector Machine (SVM) classifier. Experimental results based on the simulated data and measured data are presented, including the performance analysis for low signal-to-noise ratio (SNR) case, generalization evaluation for different target circumstances and comparison with some related methods.**

***Index Terms*—Empirical mode decomposition (EMD), feature extraction, hierarchical classification, micro-Doppler, radar target classification.**

I. Introduction

A WHEELED vehicle usually has light weight and high mobility, which is suitable for transportation task. Compared with a wheeled vehicle, a tracked vehicle has heavier weight, higher attack and defense abilities, which usually undertakes attack task. The characteristics of these two kinds of vehicles determine they have different threat degrees, thus it is important to discriminate the two kinds of vehicles for a successful strategy in modern battlefield. There have been some related work [1], [2] in this area. Recently, a special phenomenon of Doppler modulation, called micro-Doppler effect, has been introduced in radar community [3]–[8], and it could be utilized to identify different radar targets.

When radar illuminates a moving target, the carrier frequency of the returned signal will be shifted. This phenomenon is known as Doppler effect. If the target or any structural component of the target has oscillatory or rotation motions, these motions may induce additional Doppler modulations that generate sideband frequencies around the target's Doppler frequency [3], which is called as micro-Doppler effect [4], [5]. The early studies on micro-Doppler modulation focused on aeroplane and referred to the modulation as Jet Engine Modulation (JEM) [6]. Chen *et al.*[7], [8] analyzed the micro-Doppler modulation with a mathematical model in detail and introduced the micro-Doppler effect in radar. Different kinds of targets may have different kinds of micro motions, which makes micro motion, depicted by micro-Doppler modulation, a promising unique feature for identifying different kinds of radar targets. Lots of researches on the applications of micro-Doppler have been presented since the notion of micro-Doppler was induced in radar [9]–[17]. After analyzing the measured data of helicopter and human via wavelet transform and time-frequency analysis method, Thayaparan *et al.*[11] showed that different micro motions generated different micro-Doppler signatures, which makes micro motion feature extraction from micro-Doppler signal feasible. Analysis and extraction of micro-Doppler feature for pedestrians were discussed in [12]–[15]. Support Vector Machine (SVM) and Artificial Neural Network (ANN) were utilized to identify human activities via micro motion features in [13] and [14], respectively, and the results showed that human micro motion such as the motion of arms and legs could be utilized to identify human activities including running, walking, crawling and so on. References mentioned above demonstrate micro-Doppler signals and the corresponding features can be utilized for the classification of targets with different micro motions.

However, most of the previous methods are based on time-frequency analysis, which generally requires long dwell time, thus they are infeasible for surveillance radar to accomplish the task of simultaneous search and classification. When a moving vehicle is illuminated by radar within limited dwell time, a Coherent Processing Interval (CPI) is not long enough to cover the period of a micro motion structure. Consequently, it is difficult to extract effective features, such as the micro motion parameters, from the returned signal. As an alternative, the amounts of micro-Doppler components contained in the returned signals can be utilized to discriminate different micro motion targets [17], [18]. A Bayesian approach to distinguish

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human and vehicles within short dwell time was proposed in [17]. The number of micro-Doppler signals contained in the returned signal was used as the discriminative feature, which was extracted by searching the frequency peak in the time-frequency domain. Despite the limitation of short dwell time, delightful classification accuracy was achieved in [17]. A Doppler spectrum based method for vehicle classification was discussed in [18]. Based on the observation that a wheeled vehicle generally has clean Doppler spectrum and a tracked vehicle generally has much more complex Doppler spectrum, Principal Components Analysis (PCA) was used as the feature extraction technique. The notion of micro-Doppler was not mentioned in [18], however, the PCA technique was actually an approach to depict the amounts of micro-Doppler components contained in the Doppler spectrum. The classification result was finally given by the output of Linear Discriminant Analysis (LDA). Although the amounts of micro-Doppler components are different for wheeled and tracked vehicles, we desire to extract some features more directly related to the detailed structure of micro-Doppler signatures from moving vehicles.

In this paper, we develop an Empirical Mode Decomposition (EMD) based hierarchical classification structure to distinguish wheeled and tracked vehicles by using the returned signals of narrow band radar within short dwell time. Here, the detailed structure of micro-Doppler signatures from moving wheeled and tracked vehicles can be well described via EMD. The main characteristics of the proposed method can be summarized as follows.

- *EMD is utilized to decompose the motion components in the time domain signal from a moving vehicle*. Unlike Fourier transform and wavelet transform which map the signal space onto a space spanned by a predefined basis, EMD is an adaptive analysis tool that decomposes a given signal into a number of "basis functions" termed Intrinsic Mode Functions (IMFs), which are derived directly from the signal. Since EMD was first proposed by Huang *et al.* [19] in 1998, it has been widely studied by many scholars in theory [20]–[23], and in application of radar [24]–[26]. The research work in [24]–[26] showed that EMD can separate the bulk motion and micro motion in radar signals from aircrafts or vehicles. In this paper, EMD is utilized for decomposing the more detailed motion components of moving vehicles, which can be used for further classification task.
- *Based on the decomposition results of EMD, an efficient hierarchical classification structure is developed*. Since the velocity of the upper track is always twice as large as the bulk velocity, which is determined by the special structure of track, this unique feature of tracked vehicle is utilized in the first stage of our hierarchical structure to elementarily identify the tracked vehicle data. Then, if the frequency induced by the upper track does not exist from some observation aspect-sectors, the micro-Doppler components are further characterized as the discriminative feature for the two kinds of vehicles in the second classification stage.

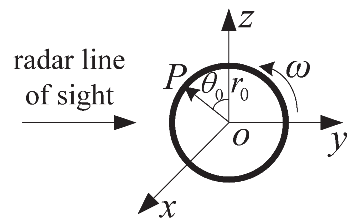


Fig. 1. Micro motion of the wheel.

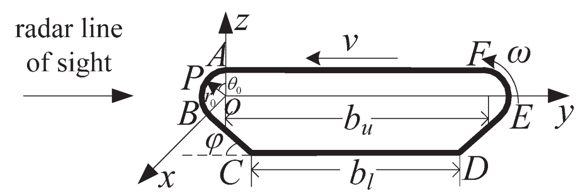


Fig. 2. Micro motion of the track.

The rest of the paper is organized as follows. In Section II, we establish the micro motion model for wheel and track. Section III analyzes the simulated data via EMD and Section IV proposes a hierarchical classification structure. The classification results and corresponding analysis based on the simulated and measured data are given in Section V, followed by the conclusion in Section VI.

II. Micro Motion Model for Wheel and Track

When a target moves along the radar line of sight (LOS), the Doppler frequency of its returned signal is

$$f_d = \frac{2v}{\lambda} \tag{1}$$

where λ is wavelength and v is radial bulk velocity. From (1), when λ is small, the Doppler frequency shift will be sensitive to the change of radial velocity. If λ is far smaller than target size, the target scattering characteristic satisfies optics assumption, and if we only consider direct scattering, the Radar Cross Section (RCS) of target can be approximately described by scattering center model [27].

In the following discussion, a local coordinate is introduced to describe the micro motion of a wheel or track, as shown in Figs. 1 and 2. In our analysis, the local coordinate has the same translation as the vehicle target, which means that the radial bulk velocity of the vehicle v has been compensated in the following micro motion analysis. First, the rotation of a single scattering point P is considered. Assume that the radius of rotation is r_0, angular velocity is ω and initial angle is θ_0. Then the returned micro motion signal of P is

$$s(t) = \xi \exp\left[j\beta \sin(\omega t + \theta_0)\right], \quad \text{with} \quad \beta = \frac{4\pi r_0}{\lambda}. \tag{2}$$

where ξ is the scattering coefficient, and t denotes time [11]. The (2) can be expanded into Fourier series

$$s(t) = \xi \sum_{n=-\infty}^{\infty} c_n e^{jn\omega t} \tag{3}$$

where the Fourier coefficient c_n is

$$c_n = \frac{\omega}{2\pi} \int_{-\pi/\omega}^{\frac{\pi}{\omega}} e^{j\beta \sin(\omega t + \theta_0)} e^{-jn\omega t} dt = e^{jn\theta_0} J_n(\beta) \tag{4}$$

with J_n denoting the nth-order Bessel function of the first kind. Substituting (4) into (3), we get the returned micro motion signal of P

$$s(t) = \xi \sum_{n=-\infty}^{\infty} e^{jn\theta_0} J_n(\beta) e^{jn\omega t}. \tag{5}$$

Then we consider the case of multi-scattering points. For a moving wheel as shown in Fig. 1, it lies in the yoz plane. Assume that K scattering points distribute evenly on the wheeled edge and the kth scattering point has an initial angle

$$\theta_k = \theta_0 + 2\pi(k-1)/K \tag{6}$$

with $k = 1, 2, \ldots, K$. The returned micro motion signal of the wheel is represented as

$$s(t) = \sum_{k=1}^{K} \xi_k \sum_{n=-\infty}^{\infty} e^{jn\theta_k} J_n(\beta) e^{jn\omega t} \tag{7}$$

where ξ_k is the scattering coefficient of the kth scattering point.

For a moving track as shown in Fig. 2, it lies in the yoz plane. The returned micro motion signal of the track is equal to the sum of the returned signals generated by parts AB, BC, CD, DE, EF and FA. The micro motions of parts AB and EF are rotation, and the returned signal of part EF is added a phase relative to that of part AB, which is caused by the length of the upper track denoted by b_u. The micro motions of parts BC, CD, DE and FA are translation. Based on the assumption that the bulk velocity is v, the rotation angular velocities of parts AB and EF with respect to the body of tracked vehicle are $\omega = v/r_0$, and the velocity of parts BC, CD, DE and FA with respect to the body are $-v\cos\varphi$, $-v$, $-v\cos\varphi$ and v, respectively, where φ is the oblique angle of the track. Consequently, the returned micro motion signal of the track is

$$\begin{aligned} s(t) &= s_{AB}(t) + s_{BC}(t) + s_{CD}(t) + s_{DE}(t) + s_{EF}(t) + s_{FA}(t) \\ &= \sum_{k=1}^{K_{\mathrm{i}}} \xi_k \sum_{n=-\infty}^{\infty} e^{jn\theta_k^{\mathrm{i}}} J_n(\beta) e^{jn\omega t} \\ &\quad + \sum_{k=1}^{K_{\mathrm{ii}}} \xi_k \exp\left[-j\frac{4\pi}{\lambda}\left(y_k^{\mathrm{ii}} + vt\cos\varphi\right)\right] \\ &\quad + \sum_{k=1}^{K_{\mathrm{iii}}} \xi_k \exp\left[-j\frac{4\pi}{\lambda}\left(y_k^{\mathrm{iii}} + vt\right)\right] \\ &\quad + \sum_{k=1}^{K_{\mathrm{iv}}} \xi_k \exp\left[-j\frac{4\pi}{\lambda}\left(y_k^{\mathrm{iv}} + vt\cos\varphi\right)\right] \\ &\quad + e^{j\frac{4\pi b_u}{\lambda}} \sum_{k=1}^{K_{\mathrm{v}}} \xi_k \sum_{n=-\infty}^{\infty} e^{jn\theta_k^{\mathrm{v}}} J_n(\beta) e^{jn\omega t} \\ &\quad + \sum_{k=1}^{K_{\mathrm{vi}}} \xi_k \exp\left[-j\frac{4\pi}{\lambda}\left(y_k^{\mathrm{vi}} - vt\right)\right] \end{aligned} \tag{8}$$

where K_{i}, K_{ii}, K_{iii}, K_{iv}, K_{v} and K_{vi} denote the scattering point numbers of parts AB, BC, CD, DE, EF and FA, respectively. θ_k^u with $u = \mathrm{i}, \mathrm{v}$ denotes the initial angle of the kth evenly distributed scattering point from AB or EF, respectively and y_k^u with $u = \mathrm{ii}, \mathrm{iii}, \mathrm{iv}, \mathrm{vi}$ denotes the initial location of the kth evenly distributed scattering point from BC, CD, DE or FA, respectively.

When bulk motion is considered, the bulk motion component with phase term $e^{j(4\pi/\lambda)vt}$, which is induced by the bulk velocity v, should be added to the returned micro-Doppler echo described in (7) or (8). Since the phase history of pulses in a CPI contains Doppler information, we can get the Doppler spectrum by applying Fourier transform to these complex pulses

$$\boldsymbol{S} = |\mathrm{fft}\,[s(t)]| \tag{9}$$

where function fft$[\cdot]$ denotes the fast Fourier transform (FFT) and operator $|\cdot|$ represents getting the complex modulus. These complex pluses in a CPI are referred to as a frame in this paper.

Considering that the material of a wheel is rubber which hardly scatters the electromagnetic wave in practice, the Doppler spectrum of the wheeled vehicle has a simple structure which mainly contains a monophonic component induced by bulk motion and very small micro-Doppler components distributed between the frequency bins representing 0 and $2v$ according to (7). The components of the Doppler spectrum of the tracked vehicle can be analyzed according to (8). The unique structure of the track determines that the velocity of the upper track, i.e., part FA, is always twice as large as the bulk velocity of the mainbody. Thus when the bulk velocity of the vehicle is v, part CD, i.e., the lower track, and part FA induce micro-Doppler components locating at the frequency bins representing 0 and $2v$, respectively, and parts AB, BC, DE and EF of the track induce micro-Doppler components distributed between the frequency bins representing 0 and $2v$. In this paper, micro-Doppler component induced by the upper track is referred to as $2v$- component. It should be emphasized that the $2v$- component cannot be always observed due to the shelter of part FA in practice.

III. Empirical Mode Decomposition and Its Application to Simulated Data

A. Review of the Empirical Mode Decomposition

Unlike Fourier transform and wavelet transform, which map the signal space onto a space spanned by a predefined basis, Empirical Mode Decomposition (EMD) is an adaptive analysis tool that decomposes a given signal into a number of "basis functions" which are derived directly from the signal [19]. Component decomposed by EMD is known as Intrinsic Mode Function (IMF) [19]. Given a signal $\boldsymbol{s}$, the first IMF can be

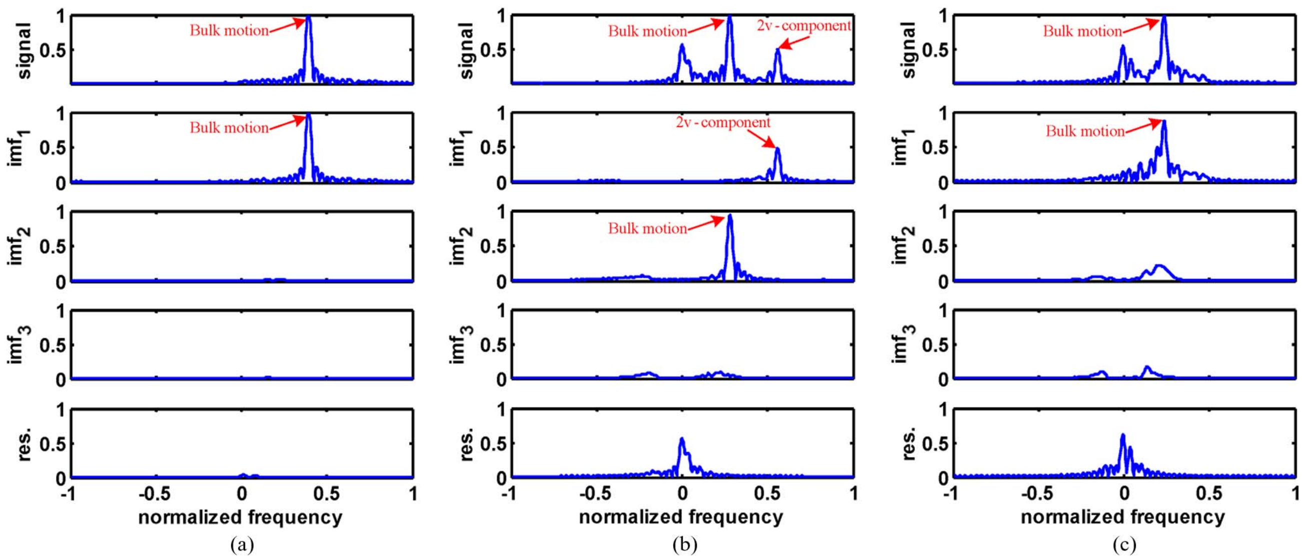


Fig. 3. Doppler spectra of original signal, IMFs and residual for simulated vehicle data. (a) Wheeled vehicle. (b) Tracked vehicle with $2v$- component. (c) Tracked vehicle without $2v$- component.

obtained in a systematic way, which is referred to as the sifting process:

(i) Let $\tilde{\boldsymbol{s}} = \boldsymbol{s}$.
(ii) Identify the local maxima of $\tilde{\boldsymbol{s}}$, and interpolate between local maxima to construct an upper envelope $\tilde{\boldsymbol{s}}_{\boldsymbol{max}}$.
(iii) Identify the local minima of $\tilde{\boldsymbol{s}}$, and interpolate between local minima to construct a lower envelope $\tilde{\boldsymbol{s}}_{\boldsymbol{min}}$.
(iv) Compute the mean values of upper and lower envelopes as $\tilde{\boldsymbol{s}}_{mean} = (\tilde{\boldsymbol{s}}_{\boldsymbol{max}} + \tilde{\boldsymbol{s}}_{\boldsymbol{min}})/2$.
(v) Extract the detail from $\tilde{\boldsymbol{s}} : \boldsymbol{m} = \tilde{\boldsymbol{s}} - \tilde{\boldsymbol{s}}_{mean}$.
(vi) Let $\tilde{\boldsymbol{s}} = \boldsymbol{m}$, and repeatedly apply steps (ii)–(v) to $\tilde{\boldsymbol{s}}$, until $\boldsymbol{m}$ is an IMF.

Once the first IMF is obtained, we store the residual $\boldsymbol{q} = \boldsymbol{s} - \boldsymbol{m}$ as a new signal, and the next IMF can be obtained by applying the sifting process to the residual $\boldsymbol{q}$. In the same way, the remaining IMFs can be calculated and EMD ends up with a representation of the form:

$$\boldsymbol{s} = \sum_{\alpha=1}^{L} \boldsymbol{m}_{\alpha} + \boldsymbol{q_L} \tag{10}$$

where $\boldsymbol{m}_\alpha$ denotes the αth IMF component and $\boldsymbol{q_L}$ is the residual after L iterations.

The original EMD can only be applied to univariate (real-valued) signals, which is not suitable for some areas using complex-valued signal structures such as sonar and radar. In [20], Rilling *et al.* propose an approach to extend EMD from a univariate case to a bivariate case. The main idea of this approach is that it considers the real-valued signal as the scale of its local oscillations, which means fast oscillations superimpose on slow oscillations, and the complex-valued signal as the scale of its local rotations, which, similar to the real-valued case, means fast rotations superimpose on slow rotations. After the viewpoint for complex-valued signal is established, the bivariate EMD can be achieved as long as the slow rotation component is defined by the mean of some "envelope," and the two definitions of the mean processing can be found in [20].

To understand the behavior of EMD as a decomposition procedure, Flandrin *et al.* [22] show that EMD acts essentially as a filter bank resembling those involved in wavelet decompositions. In this view, the filter associated to the first IMF is essentially high-pass, and the IMFs of higher indices are characterized by a set of overlapping bandpass filters. Moreover, each mode with index $= (\alpha + 1)$, $\alpha \geq 2$, occupies a frequency domain which is roughly the upper half band of that of the previous residual with index $= \alpha$. Though IMFs and residual can intuitively be given a "spectral" interpretation, it is worth stressing the fact that they corresponds rather an automatic and adaptive time-variant filtering than a pre-determined subband filtering. Consequently, EMD can adaptively filter different narrowband frequency signals without predetermined basis for a given signal.

B. Empirical Mode Decomposition of the Simulated Data

For a frame of slow time signal, we can get its EMD results as (10), and then the Doppler spectra of each IMF and the residual can be computed according to (9).

The example Doppler spectra of IMFs, residual and original signal for the simulated data are shown in Fig. 3. Here, the iteration number, namely L, is 3 for comparison convenience. The decomposition results for the wheeled vehicle are shown in Fig. 3(a). As we mentioned in Section II, the micro-Doppler components of the wheel are weak, therefore, the Doppler spectra of IMFs are very small except for that of the first IMF, namely $\boldsymbol{m}_1$, which mainly contains the bulk motion component. The results for the tracked vehicle with $2v$- component and without $2v$- component are given in Fig. 3(b) and (c), respectively. When the $2v$- component is observed as in Fig. 3(b), it can be captured by $\boldsymbol{m}_1$, the bulk motion component is contained in the second IMF namely $\boldsymbol{m}_2$, and the rest

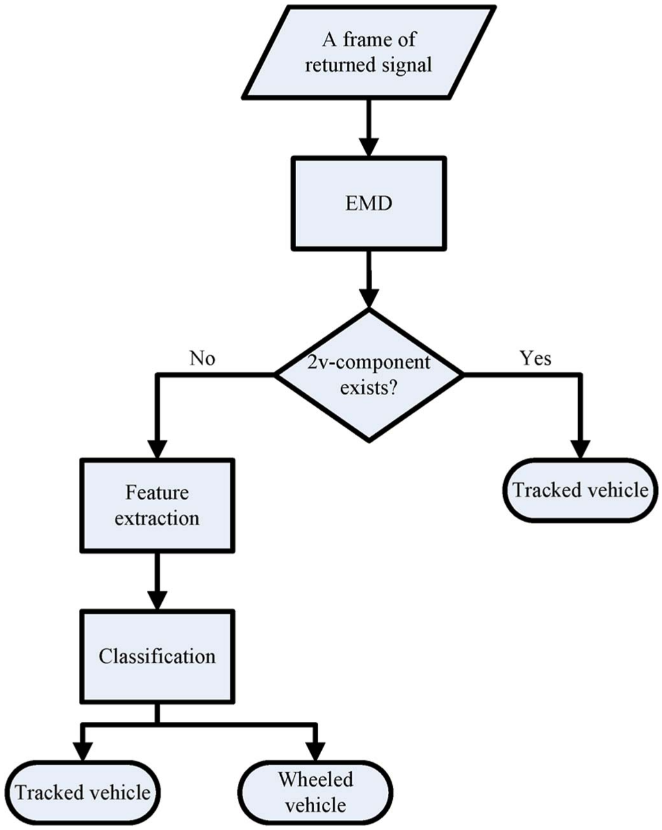


Fig. 4. Hierarchical classification structure for moving vehicles.

micro-Doppler components are contained in the third IMF namely $\boldsymbol{m}_3$. When the $2v$- component is invisible as in Fig. 3(c), $\boldsymbol{m}_1$ mainly contains the bulk motion component, and the rest of IMFs, namely $\boldsymbol{m}_2$ and $\boldsymbol{m}_3$, contain the micro-Doppler components. In addition, the micro-Doppler component induced by the lower track is contained in the residual for the tracked vehicle.

IV. Hierarchical Classification Structure

From the analysis of the simulated data, we learn that the performance of EMD in decomposition process is similar to an adaptive filter bank. Due to such good characteristic of EMD, when the $2v$- component induced by upper track can be observed, the first IMF captures the $2v$- component, and when the $2v$- component cannot be observed, the first IMF contains the bulk motion component. This fact, which is actually the result of the adaptive filter character of EMD, means we can utilize the first IMF as an indicator to check whether the $2v$- component presents. Thus a hierarchical classification structure which is composed of two stages is proposed as shown in Fig. 4. When we obtain the decomposition results of a frame of the returned signal via EMD, The existence of $2v$- component will be checked in the first stage of the hierarchical structure and an elementary identification result for the tracked vehicle can be given since $2v$- component is the unique feature of the tracked vehicle determined by its structure. Considering that micro-Doppler signatures are also dependent on the target-radar orientation, the $2v$- component may not be observed from some observation aspect sector. In this situation, some features, which can depict the detailed distribution structure of Doppler spectra of IMFs and the relation between IMFs, will be extracted from the decomposition results of the returned signals and a further discrimination will be performed by the selected state-of-art classifier for the two kinds of vehicles in the second classification stage.

To realize the two classification stages, we firstly define two symbols. After applying EMD to the slow time signal, we can get the decomposition results as (10). Define $\boldsymbol{M}_1$ and $\boldsymbol{M}_r$ as follows:

$$\boldsymbol{M}_1 = |\text{fft}[\boldsymbol{m}_1]| \tag{11}$$

$$\boldsymbol{M}_r = \left|\text{fft}\left[\sum_{\alpha=2}^{L} \boldsymbol{m}_\alpha\right]\right|. \tag{12}$$

A. First Stage of the Hierarchical Structure

In the first stage, our goal is to check the existence of the $2v$- component. An example $\boldsymbol{M}_1$ and $\boldsymbol{M}_r$ of the two kinds of vehicles are given in Fig. 5. As in Fig. 5(a), the $\boldsymbol{M}_1$ of the wheeled vehicle mainly contains bulk motion component while $\boldsymbol{M}_r$, which contains micro motion components, is very weak. For the tracked vehicle, when the $2v$- component can be observed as in Fig. 5(b), $\boldsymbol{M}_1$ captures $2v$- component while $\boldsymbol{M}_r$ contains bulk motion and micro motion components; and when the $2v$- component cannot be observed as in Fig. 5(c), $\boldsymbol{M}_1$ mainly contains bulk motion component while $\boldsymbol{M}_r$ contains micro motion components. Based on this observation, if the $2v$- component presents, then

(i) $|\text{position}[\boldsymbol{M}_1] - 2 \times \text{position}[\boldsymbol{M}_r]| < \varepsilon$, with ε being a small positive number
(ii) $\text{energy}[\boldsymbol{M}_1] < \text{energy}[\boldsymbol{M}_r]$
(iii) $\max[\boldsymbol{M}_1] < \max[\boldsymbol{M}_r]$

where, for a series $\boldsymbol{M} = \{M(i)\}_{i=1}^N$, position$[\cdot]$, energy$[\cdot]$ and $\max[\cdot]$ are defined as follows:

$$\text{position}[\boldsymbol{M}] = \arg_i \left(\max\left(\{M(i)\}_{i=1}^N\right)\right) - \frac{N}{2} - 1 \tag{13}$$

$$\text{energy}[\boldsymbol{M}] = \sum_{i=1}^{N} M(i)^2 \tag{14}$$

$$\max[\boldsymbol{M}] = \max\left(\{M(i)\}_{i=1}^N\right) \tag{15}$$

and ε in condition (i) is an error-controlling constant which is usually a small positive number. In these three conditions, condition (i) describes the velocity (or Doppler frequency) relationship between $\boldsymbol{M}_1$ and $\boldsymbol{M}_r$; while conditions (ii) and (iii) are proposed to avoid the situation that the Doppler frequencies locating at v and $v/2$ satisfy condition (i), which is based on the observation that the bulk motion component is maximum in the Doppler spectrum. If and only if all the three conditions hold, we can affirm the existence of the $2v$- component and the target can be identified as tracked vehicle.

B. Second Stage of the Hierarchical Structure

As to a frame that does not contain $2v$- component, from Fig. 5(a) and (c) we find that the micro-Doppler components

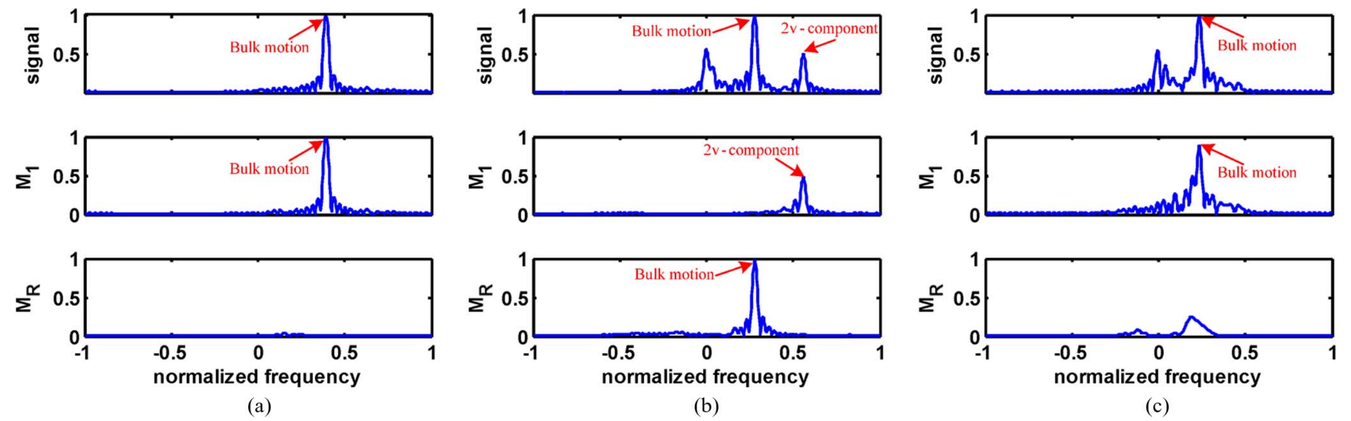


Fig. 5. $\boldsymbol{M}_1, \boldsymbol{M}_r$ and Doppler spectra of the simulated signal for moving vehicles. (a) Wheeled vehicle. (b) Tracked vehicle with $2v$- component. (c) Tracked vehicle without $2v$- component.

contained in $\boldsymbol{M}_1$ and $\boldsymbol{M}_r$ of the wheeled vehicle are small while those of the tracked vehicle are large. Therefore, the following five features can be exploited to depict the distinction between tracked vehicle and wheeled vehicle:

The first feature

$$\text{feature1} = \frac{\max[\boldsymbol{M}_r]}{\max[\boldsymbol{M}_1]}. \tag{16}$$

The second feature

$$\text{feature2} = \frac{\text{energy}[\boldsymbol{M}_r]}{\text{energy}[\boldsymbol{M}_1]}. \tag{17}$$

The third feature

$$\text{feature3} = \text{entropy}[\boldsymbol{M}_1] \tag{18}$$

where entropy$[\cdot]$ is defined for a series $\boldsymbol{M} = \{M(i)\}_{i=1}^{N}$ as follows:

$$\text{entropy}[\boldsymbol{M}] = -\sum_{i=1}^{N} p_i \ln p_i \tag{19}$$

where $p_i = M(i)/\sum_{i=1}^{N} M(i)$. The entropy indicates the energy distribution of a series, the more the energy gathers, which means the more a series is close to an impulse function, the less the entropy is. It should be noted that this feature is sensitive to the width of the Doppler spectrum, thus the spectrum need to be normalized by a reference velocity so that all the spectra have the same width before this entropy feature is extracted.

The fourth feature

$$\text{feature4} = \frac{\max\left[\text{clean}[\boldsymbol{M}_1]\right]}{\max\left[\text{clean\_r}[\boldsymbol{M}_1]\right]} \tag{20}$$

where the operations clean$[\cdot]$ and clean_r$[\cdot]$ are based on the CLEAN technique, which can extract harmonic from a given signal [28]. In $\boldsymbol{M}_1$ from the measured vehicle data without $2v$- component, the maximum amplitude should correspond to the bulk motion component. Therefore, applying CLEAN technique to $\boldsymbol{M}_1$ once, we can get a monophonic frequency signal and a residual signal, of which the monophonic frequency signal corresponding to the bulk motion component is designated as the result of clean$[\cdot]$, and the residual signal corresponding to the micro motion components is designated as the result of clean_r$[\cdot]$.

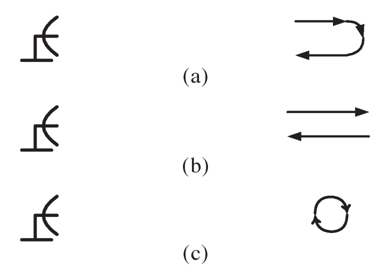


Fig. 6. Three scenarios of measured data collection. (a) Vehicle first recedes from radar, then turns around and approaches radar. (For both wheeled vehicle and tracked vehicle) (b) Vehicle recedes from radar and approaches radar. (For both wheeled vehicle and tracked vehicle) (c) Vehicle circles around. (For tracked vehicle only).

The fifth feature

$$\text{feature5} = \frac{\text{energy}\left[\text{clean}[\boldsymbol{M}_1]\right]}{\text{energy}\left[\text{clean\_r}[\boldsymbol{M}_1]\right]}. \tag{21}$$

The above five features are extracted from the spectra of IMFs, another feature that depicts energy relationship between IMFs can be directly extracted from IMFs

$$\text{feature6} = \arg_{\eta}\left(\sum_{\alpha=1}^{\eta}\text{energy}[\boldsymbol{m}_\alpha]/\sum_{\alpha=1}^{L}\text{energy}[\boldsymbol{m}_\alpha] \geq 95\%\right) \tag{22}$$

where L is the number of IMFs.

The meaning of these features is given as follows:

- Features 1 and 2 describe the relationship between $\boldsymbol{M}_r$ and $\boldsymbol{M}_1$. As shown in Fig. 5(a) and (c), the micro-Doppler components of the wheeled vehicle are weak and those of the tracked vehicle are remarkable. Therefore, the peak value ratio and energy ratio of the wheeled vehicle between $\boldsymbol{M}_r$ and $\boldsymbol{M}_1$ are very small, while those of the tracked vehicle are relatively large.
- Features 3, 4 and 5 describe the properties of $\boldsymbol{M}_1$. The returned signal of the wheeled vehicle is approximately monophonic, thus its $\boldsymbol{M}_1$ has concentrative energy which means small entropy, high bulk-to-micro peak ratio and high bulk-to-micro energy ratio as shown in Fig. 5(a). In

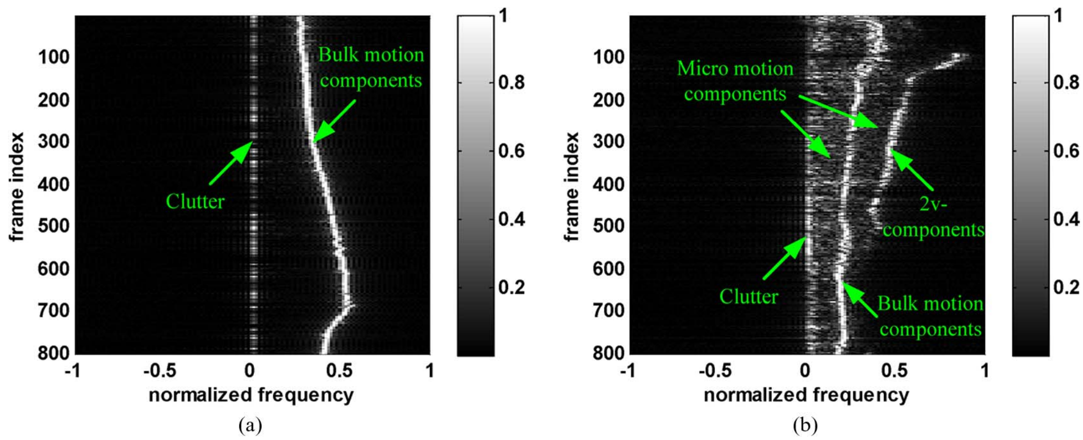


Fig. 7. Doppler spectrum groups of the moving vehicles. (a) Wheeled vehicle. (b) Tracked vehicle.

contrast, the returned signal of the tracked vehicle contains remarkable micro-Doppler components, thus its M_1 has large entropy, low bulk-to-micro peak ratio and low bulk-to-micro energy ratio as in shown Fig. 5(c).

- Feature 6 describes the normalized energy distribution between IMFs. To reach the ratio 95%, more IMFs are needed for the tracked vehicle which has remarkable micro-Doppler components than the wheeled vehicle which is approximately monophonic.

V. Experiment Results

A. Introduction and Analysis of the Measured Data

Measured data are collected from one wheeled vehicle and one tracked vehicle under three experimental scenarios, which are illustrated in Fig. 6. Radar scans from a fixed position when it collects the data of the target and only a single vehicle target is tested at one time. In scenario (a), the vehicle first recedes from radar, then turns around and approaches radar. Radar collects the returned signals of targets without interruption during the movement of vehicles. In scenario (b), radar records the measured data of a vehicle twice. First, the vehicle recedes from radar, and secondly the vehicle approaches radar. The mesured data are collected for both wheeled and tracked vehicles according to scenarios (a) and (b). Radar also collects the data only for the tracked vehicle which moves in a circle according to scenario (c). All the measured data are collected in high Signal-Noise-Ratio (SNR) condition.

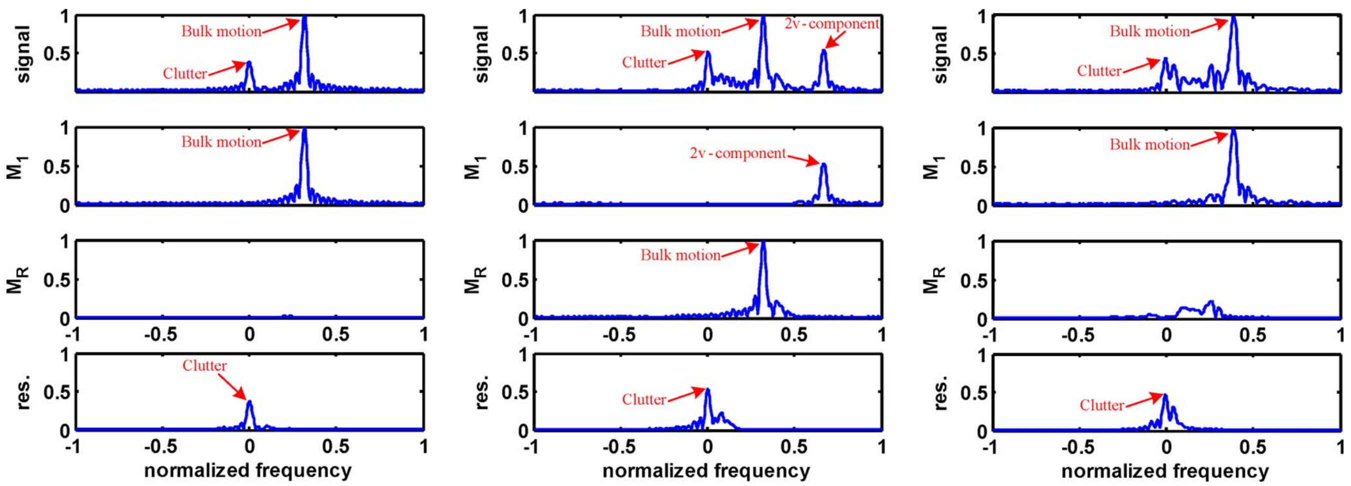
The Doppler spectrum groups of the wheeled and tracked vehicles without clutter suppression are illustrated in Fig. 7(a) and (b), respectively, where the horizontal axis represents normalized frequency, the vertical axis represents frame index and the different gray-scale values represent the amplitudes of Doppler spectra. The wheeled vehicle has monophonic frequency spectra, and the entire Doppler spectrum plane is dominated by the bulk motion components and clutter as shown in Fig. 7(a). For the tracked vehicle, the remarkable micro-Doppler components can be observed in the Doppler plane as shown in Fig. 7(b). We can see the $2v$- components only exist in the Doppler spectra of the frames indexed from 100 to 500, which means the $2v$- component cannot be always observed. It should be emphasized that the Doppler frequency induced by the lower track has mixed with clutter for the tracked vehicle, thus this micro-Doppler information cannot be utilized for classification.

The analysis of frames of measured data from the wheeled and tracked vehicles via EMD is shown in Fig. 8. As we see that the decomposition results of measured data are similar to those of simulated data discussed in Sections III-B and IV-A except that the clutter is contained in the residual. The first IMF can be utilized to check the existence of the $2v$- component. It should be noticed that the residual cannot be used to extract useful features due to the existence of clutter. However, clutter suppression can be achieved by discarding the residual without extra preprocessing. It is another advantage of EMD in its application to the measured data.

B. Classification Results

As we mentioned in Section IV, the measured data to be tested are checked in the first stage of the hierarchical classification structure to elementarily identify the tracked vehicle according to the existence of the $2v$- component. We search for the frames with $2v$- component across all measured data by checking whether each frame satisfies all the three conditions of the first stage proposed in Section IV. The statistic results are given in Table I. From the experiment, there are 19.18% frames with $2v$- component for the tracked vehicle data; while the proportion is 0.08% for the wheeled vehicle data, which is very low relative to that of the tracked vehicle. We select the frames with $2v$- component in the wheeled vehicle data for further analysis, and find that the corresponding radial velocities are very small. This indicates the wheeled vehicle has tangential movement relative to radar, and its Doppler spectra in this condition become complicated. The statistic results in Table I demonstrate that the $2v$- component is a distinct feature for discriminating the wheeled and tracked vehicles.

For those frames without $2v$- component, a classification process needs to be implemented in the second stage of the hierarchical structure. When we get the results of the first and second stages, the final classification results can be obtained as follows. Assume that there are total D_w frames' data from the wheeled vehicle in the test data, of which D_{w1} frames satisfies all the three conditions of the first stage proposed in Section IV



(a) (b) (c)

Fig. 8. $\boldsymbol{M}_1, \boldsymbol{M}_r$ and Doppler spectra of original signal and residual for vehicles from measured data. (a) Wheeled vehicle. (b) Tracked vehicle with $2v$- component. (c) Tracked vehicle without $2v$- component.

TABLE I
PROPORTION OF FRAMES WITH $2v$- COMPONENT IN THE MEASURED DATA

	Wheeled	*Tracked*
Total number of frames	6003	5885
Number of frames with 2v- component	5	1129
Proportion	0.08%	19.18%

and the proportion of these frames is $C_{w1} = D_{w1}/D_w$ (this proportion is given by the first stage as in Table I), thus the total number of frames used in the second stage are $D_w - D_{w1}$. If there are D_{w2} frames which are distinguished as the wheeled vehicle data in the second stage, the classification rate of the second stage will be $C_{w2} = D_{w2}/(D_w - D_{w1})$. Then the final classification rate of the wheeled vehicle is

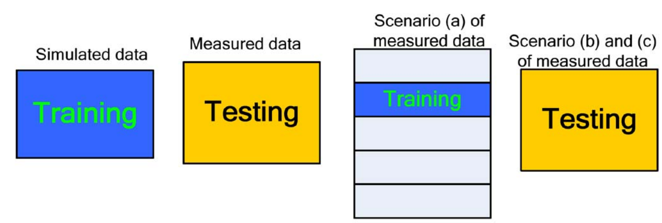
$$C_w = \frac{D_{w2}}{D_w} = \frac{(D_w - D_{w1})C_{w2}}{D_w} = (1 - C_{w1})C_{w2}. \quad (23)$$

In the similar way, the final classification rate of the tracked vehicle is

$$C_t = \frac{D_{t2} + D_{t1}}{D_t} = \frac{(D_t - D_{t1})C_{t2} + D_{t1}}{D_t} = (1 - C_{t1})C_{t2} + C_{t1} \quad (24)$$

where D_t, D_{t1}, D_{t2} are the total number of frames from the tracked vehicle in the test data, the number of frames with $2v$- component found in the first stage and the number of frames correctly identified to be the tracked vehicle data in the second stage, respectively. C_{t1} and C_{t2} denote the proportion of frames with $2v$- component for the test data from the tracked vehicle given by the first stage and the classification rate for the tracked vehicle given by the second stage, respectively.

We design two experimental schemes for the second stage of the hierarchical structure to validate the classification performance as shown in Fig. 9. In the first scheme, we employ the simulated data as training set, and the measured data of all three scenarios as test data. The parameters of vehicles in the simulation are shown in Table II.



(a) (b)

Fig. 9. Schemes for the selection of the training set and test set. (a) The first scheme, simulated data as training set and measured data as test set. (b) The second scheme, measured data as training set and measured data as test set.

TABLE II
TARGET PARAMETERS IN SIMULATION

Parameter list	*Wheel*	*Track*
Radius: r_0 (m)	0.3	0.2
Length of the upper track: b_u (m)	-	4
Length of the lower track: b_l (m)	-	2
Oblique angle: φ (rad)	-	π/8
Range of bulk velocity (m/s)	2.6-7.1	2-3.9

Support Vector Machine (SVM) [29] is chosen as classifier for the extracted features described in Section IV-B. Radial Basis Function (RBF) is selected as the kernel of SVM. For the first experimental scheme the parameters of SVM are optimized to have the minimum average classification error by searching in a certain range; while for the second experimental scheme, the parameters are determined by 5-fold cross-validation. Classification accuracy of the first experimental scheme is given in Table III. The final classification rates of the wheeled and tracked vehicles are computed according to (23) and (24), respectively. We can see the final classification rate of the wheeled vehicle, which considers the proportion of frames satisfying the three conditions in the first stage, descends very slightly compared with its classification rate of the second stage; while the final classification rate of the tracked vehicle has an obvious improvement compared with its classification rate of the second stage. The final average classification rate is 84.61%, which

TABLE III
CONFUSION MATRIX AND FINAL CLASSIFICATION RATE (FCR) OF THE FIRST EXPERIMENTAL SCHEME. THE TOTAL TEST FRAMES OF THE WHEELED AND TRACKED VEHICLES ARE 6003 AND 5885, RESPECTIVELY

		Wheeled	*Tracked*	*FCR*	*Average FCR*
Wheeled	***1st stage***	No decision	5 (0.08%)	80.07%	84.61%
	2nd stage	4806 (80.13%)	1192		
Tracked	***1st stage***	No decision	1129 (19.18%)	89.16%	
	2nd stage	638	4118 (86.59%)		

TABLE IV
CLASSIFICATION RATE OF THE SECOND EXPERIMENTAL SCHEME (%)

Trial index	*Wheeled*	*Tracked*	*Average*
1	94.20	90.56	92.38
2	92.01	90.32	91.16
3	94.38	93.42	93.90
4	93.05	89.23	91.14
5	93.00	89.69	91.35
Average of 5 trials	93.33	90.65	91.99

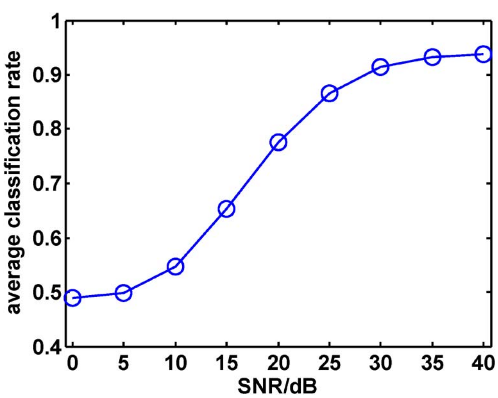


Fig. 10. Classification rate changing versus SNR.

shows that the micro motion model proposed in Section II accords with the measured data.

In the second experimental scheme, we utilize the measured data from scenario (a) as training set and those from scenario (b) and (c) as test set to evaluate the generalization performance of the proposed features under different target circumstances. Training set is uniformly divided into 5 subsets, i.e., assign the 1st, 6th, 11th, ... samples to subset 1, the 2nd, 7th, 12th, ... samples to subset 2, and so on. Total 5 trials are implemented in the second scheme. In each trial, we use one subset for training and all the test data for testing.

The final classification results of the second experimental scheme are given in Table IV. The average classification rate over the 5 trials in scheme (b) is about 92%. Although the training data and test data come from different target circumstances, which mean that the aspect, velocity and clutter may differ in different measurement scenarios, the hierarchical classification structure can discriminate the wheeled and the tracked vehicles effectively.

In practice, the classification performance in different test SNR conditions is an important evaluation for a classification

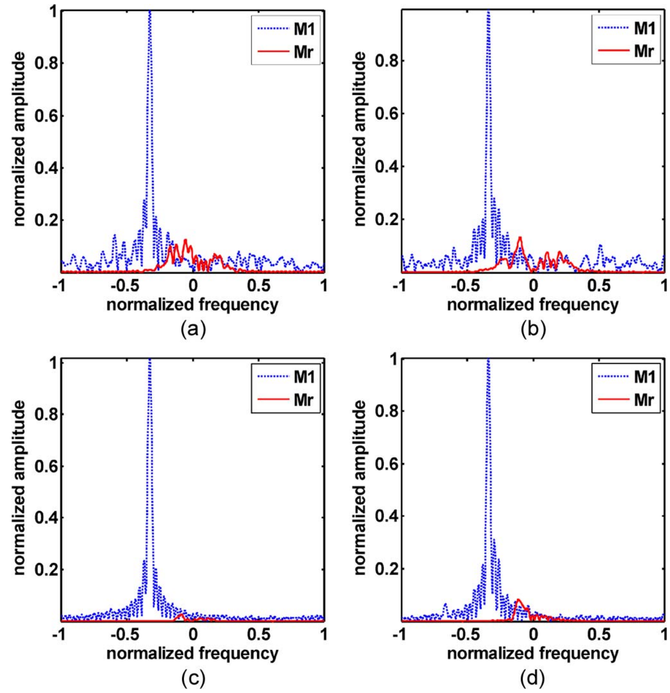


Fig. 11. M_1 and M_r for vehicles with different SNR conditions. (a) Wheeled vehicle with SNR = 10 dB. (b) Tracked vehicle with SNR = 10 dB. (c) Wheeled vehicle without adding noise (high SNR). (d) Tracked vehicle without adding noise (high SNR).

TABLE V
CLASSIFICATION RATES OF THE DIFFERENT METHODS (%)

Method List	*Wheeled*	*Tracked*	*Average*
DFT	85.28	84.39	84.84
DCT	84.61	82.00	83.30
Wavelet	82.38	93.15	87.77
PCA	91.08	81.82	86.45
EMD	93.33	90.65	91.99

TABLE VI
CHARACTERISTICS OF THE DIFFERENT METHODS

Characteristic List	*DFT*	*DCT*	*Wavelet*	*PCA*	*EMD*
Frequency structure expression	Yes	Yes	Yes	No	Yes
Bulk and 2v- components separation	No	No	Partly	No	Yes
Clutter suppression	No	No	No	No	Yes

algorithm, which is also a valuable reference for application in real world scenario. Here, we evaluate the classification performance of our method in different test SNR conditions based on scheme (b). The measured data from scenario (a) with high SNR are used as the training data, while the test data with different SNR are obtained by adding white noise to the measured data from scenario (b) and (c). In this way, we can evaluate the robustness of the proposed method in different test SNR conditions. The average classification rates of 5 independent trials according to scheme (b) versus different SNR are shown in Fig. 10. Classification rate is stable above 90% when the test SNR is between 30 dB and 40 dB, and dramatically decreases when the SNR is below 20 dB. This

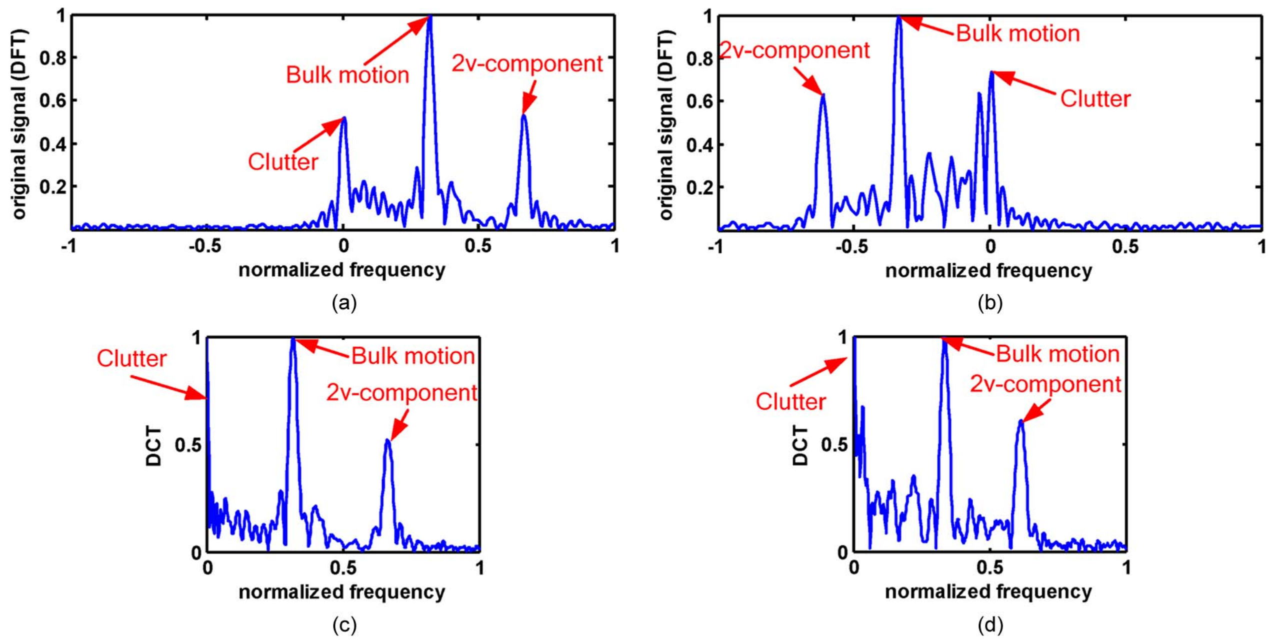


Fig. 12. Analysis results for the tracked vehicle in two target aspects. (a) DFT result when the vehicle approaches radar. (b) DFT result when the vehicle recedes from radar. (c) DCT result when the vehicle approaches radar. (d) DCT result when the vehicle recedes from radar.

can be explained by the Doppler spectra of the wheeled and tracked vehicles with SNR $= 10$ dB as shown in Fig. 11(a) and (b), respectively. In that case, except clutter, the Doppler spectrum of the wheeled vehicle no longer approximates to a monophonic component (compared with that without adding noise as shown in Fig. 11(c)). In Fig. 11(a), there are remarkable components distributing across the whole frequency axis. These new frequency components are induced by white noise, and some of them have relative large amplitude which are similar to the micro-Doppler components and could mislead the classifiers. For the Doppler spectrum of the tracked vehicle shown in Fig. 11(b), we could find a lot of components distributing between the frequency bins that represent $2v$ and 0, which are induced by not only the micro motion but also the noise, and components distributing in other frequency bins which are induced by noise (compared with that without adding noise as shown in Fig. 11(d)). Since the Doppler spectra of the two vehicles under low SNR are very like to each other, the classification performance dramatically deteriorates.

C. Comparison With Related Techniques

In this part, a comparison between EMD and some classical or state-of-art methods will be given based on our measured data set. These related techniques include discrete Fourier transform (DFT) [30], discrete cosine transform (DCT) [31], wavelet transform [32] and principal component analysis (PCA) [33]. They are all successful analysis or decomposition methods in signal processing applications and some of them have been utilized for micro-Doppler analysis and classification already [11], [18]. In these methods, DFT, DCT and wavelet transform are independent of data, i.e., their bases are predefined; while PCA is data-driven whose bases are derived directly form the data, which is similar to EMD. It should be noted that DFT of micro-Doppler signature is actually the Doppler spectrum. In the comparison experiment, we use scheme (b) to evaluate the performance of different methods. For DFT, DCT and wavelet transform, we first use them to analyze or decompose the micro-Doppler signals of moving vehicles, and then the feature extraction stage is implemented on the analysis or decomposition results. As to PCA, we employ the similar preprocessing method as in [18] but use SVM classifier instead of the linear discriminant classifier for the equitable comparison. In addition, we adapt the SVM parameters for each method by 5-fold cross-validation, and the classification result is based on their optimal parameters.

The classification results based on different analysis or decomposition methods are given in Table V. We can see EMD achieves the best classification performance. To analyze the advantages of EMD for moving vehicle classification, the comparisons of some analysis or decomposition characteristics between different methods in their application to micro-Doppler signals are listed in Table VI. The detailed analyses of these characteristics of different methods are discussed as follows.

1) DFT and DCT Case: DCT is similar to DFT except that DCT treats two symmetrical frequencies relative to 0 as the same one, and therefore DCT cannot distinguish the symmetrical frequency components. The analysis results for measured moving vehicle data via DFT, DCT are given in Fig. 12. The two frames used in Fig. 12 are selected from two different target aspects: 1) the frame with positive velocity in Fig. 12(a) indicates the vehicle approaches radar; 2) the frame with negative velocity in Fig. 12(b) indicates the vehicle recedes from radar. When the vehicle has positive velocity, the result of DCT as in Fig. 12(c) is similar to that of DFT as in Fig. 12(a). However, when the vehicle has negative velocity, DCT obtains a mirror image for negative frequency components relative to 0 as in Fig. 12(d) compare with the result of DFT as in Fig. 12(b). In practice, since the returned signals of moving vehicles have

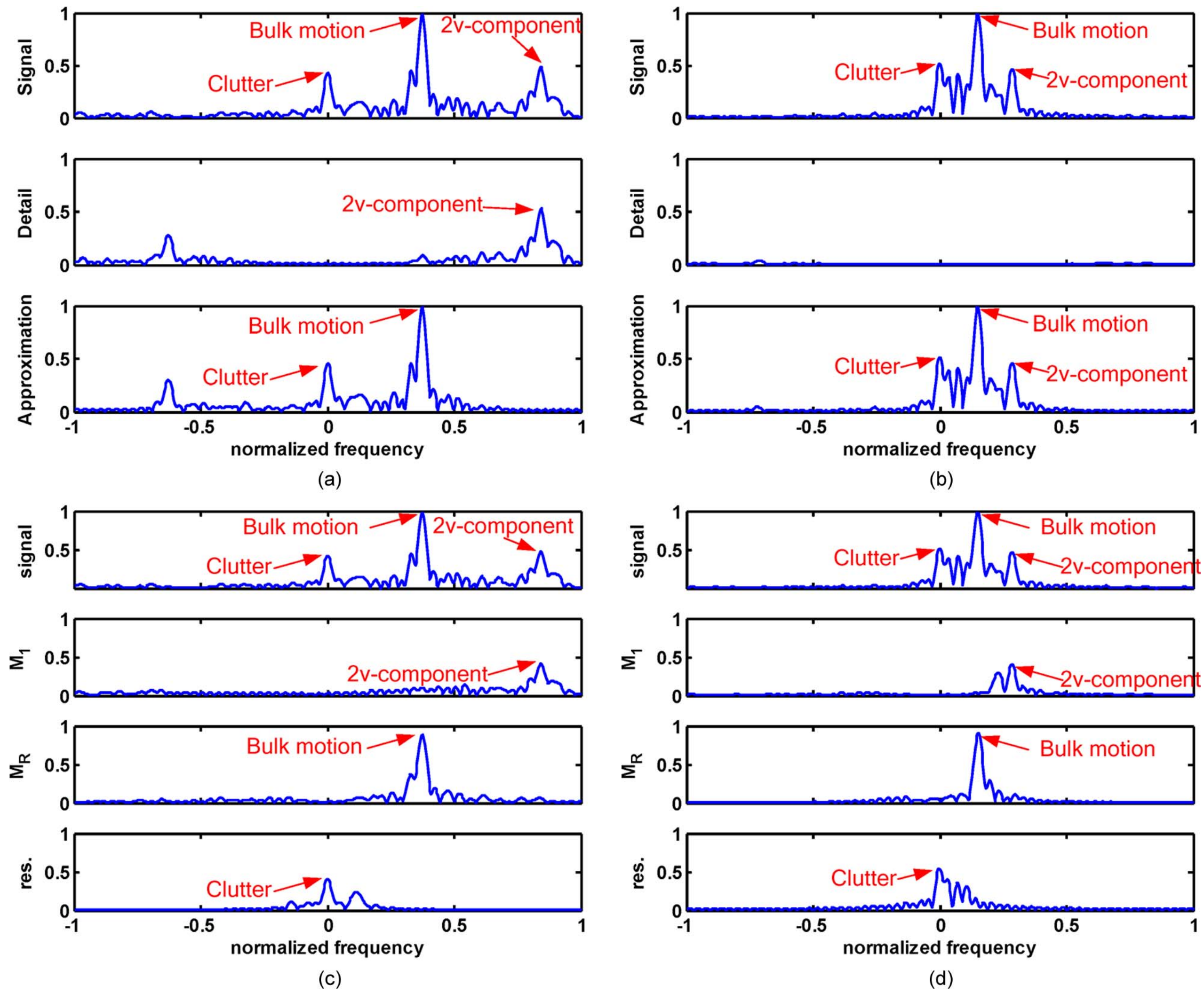


Fig. 13. Decomposition results for the tracked vehicle with the high or low bulk velocity. (a) Wavelet transform results with the high bulk velocity. (b) Wavelet transform results with the low bulk velocity. (c) EMD results with the high bulk velocity. (d) EMD results with the low bulk velocity.

unsymmetrical frequency components relative to 0, that is the frequencies induced by moving vehicles distribute in the range of $[0, 2v], (v > 0)$ as in Fig. 12(a) or $[2v, 0], (v < 0)$ as in Fig. 12(b), when we utilize DCT to decompose the returned signal, the noise distributed in the range of $[-2v, 0], (v > 0)$ or $[0, -2v], (v < 0)$ will mix with the bulk motion component and the micro-Doppler component. Therefore, the discriminative information contained in the bulk motion component or the micro-Doppler component might be contaminated by the noise component via DCT.

Additionally, in contrast with EMD, DFT and DCT are more like spectral analysis methods that can only depict different frequency components contained in a given signal, but fail to decompose the given signal into the components like the $\boldsymbol{M}_1, \boldsymbol{M}_r$ and residual of EMD, which respectively contain the $2v$- component, bulk motion component and clutter. Thus, as listed in Table VI, DFT and DCT cannot separate important frequency components such as 2v-component, bulk motion component and clutter from each other, and they cannot accomplish clutter suppression as EMD does either.

2) Wavelet Transform Case: Similar to DFT and DCT, wavelet transform is a predefined basis decomposition technique. Although wavelet transform is able to separate different frequency bands, its equivalent filter banks are pre-determined, which are not adaptive to the frequency bands of input signals. A wavelet with basis Daubechies 6 (db6) is selected to generate the high-pass and low-pass decomposition results, and the one-level wavelet decomposition components of the micro-Doppler signals of the tracked vehicle with different bulk velocities are shown in Fig. 13. When the vehicle has high bulk velocity, its Doppler spectrum occupies wide-range bins in frequency domain. In this case, the wavelet transform separates bulk motion and $2v$- components as in Fig. 13(a). The bulk motion and $2v$- components locate in the approximation component and the detail component, respectively. However, when the vehicle has low bulk velocity, its Doppler spectrum concentrates in only a few frequency bins near to 0, which means it locates in the low frequency band. In this case, most frequency components are contained in the approximation component, and therefore the bulk motion and $2v$- components are not separated as in Fig. 13(b). Thus, as listed in Table VI, wavelet transform can only separate the bulk motion and $2v$- components in some cases due to the limitation of predetermined filter banks. Unlike wavelet transform, EMD is an adaptive technique that

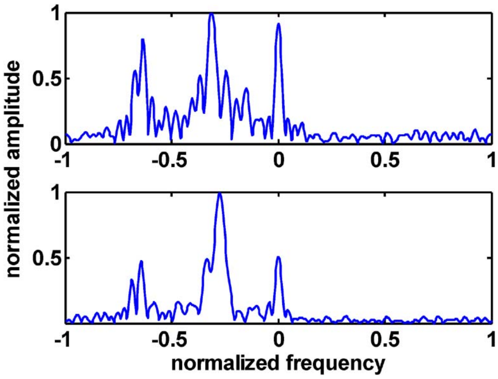


Fig. 14. Doppler spectra of two consecutive frames of the tracked vehicle.

decomposes a given signal into a number of IMFs which are derived directly from the signal. As shown in Fig. 13(c) and (d), due to the adaptive filtering characteristic, EMD can separate the bulk motion and $2v$- components no matter how large the bulk velocity is.

Additionally, since it is difficult for wavelet transform to well separate clutter from signal (even in Multilevel wavelet decomposition, due to the variation of the signal bandwidth with the bulk velocity, which level containing clutter is unknown), wavelet transform cannot accomplish clutter suppression as EMD does, which is also listed in Table VI.

Lastly, before decomposing a given signal via wavelet transform, we need to choose suitable wavelet basis. Otherwise the decomposition results may be unsatisfied. Since the predefined bases do not adaptively match the input signal, it limits the application of wavelet transform to automatic target classification task. Thus, in this sense, EMD is more effective than wavelet transform or other predefined basis methods in a real-world application.

3) PCA Case: PCA projects data into a subspace whose basis vectors correspond to the maximum-variance directions in the original space. The bases of PCA need to be orthogonal to each other, i.e., when the first basis is obtained, the next basis can only be found in the directions which are orthogonal to the first one. In practice, since the micro-Doppler signatures of moving vehicles usually change rapidly, the Doppler spectrum of one frame may be different from that of the next frame even they are from the same vehicle. An instance in Fig. 14 shows the Doppler spectra of two consecutive frames from the tracked vehicle. The amplitudes of the two consecutive Doppler spectra at some frequency bins are very different from each other. This observation indicates that the amplitudes of Doppler spectra from the same target may have large variance at some frequency bins. In this case, the maximum-variance directions found by PCA may not be the most distinguishable directions between the wheeled and tracked vehicles. Additionally, since the transformed data no longer have explicit frequency meaning in the new space, the frequency structure information contained in the Doppler spectrum will be lost after PCA. As a result, the features which can reflect structure information such as $2v$- component cannot be utilized. Therefore, as listed in Table VI, PCA does not preserve the frequency structure of the original signal, cannot distinguish different frequency components, i.e., the bulk motion and $2v$- components, and is unable to suppress clutter as EMD does.

In a few words, EMD has better decomposition performance in frequency structure expression, bulk motion component and $2v$- component separation, and clutter suppression than DFT, DCT, wavelet transform and PCA for the micro-Doppler signature analysis of the moving wheeled and tracked vehicles. These advantages make EMD a better choice for the classification task of the moving wheeled and tracked vehicles, and accordingly, satisfactory results are obtained.

VI. Conclusion

In this paper, the micro motion models of moving vehicles based on the multi-scattering centers theory are established and the Doppler spectrum structures of the wheeled and tracked vehicles are analyzed. From the analysis, the micro-Doppler component induced by the upper track can be considered as an important feature of the tracked vehicle although it cannot always be observed in practice. In addition, the micro-Doppler components contained in the returned signals are different between wheeled and tracked vehicles. According to these analyses, a hierarchical classification structure based on EMD is proposed for distinguishing the wheeled and tracked vehicles, where EMD is utilized to not only capture the $2v$- component of the tracked vehicle in the first stage but also extract distinctive features in the second stage. The proposed method can well describe the detailed structure of micro-Doppler signatures from moving wheeled and tracked vehicles in comparison with some related techniques such as DFT, DCT, wavelet transform, and PCA. From experimental results based on simulated and measured data, the proposed micro motion models for moving vehicles are validated and the proposed method obtains good classification performance under different target circumstances.

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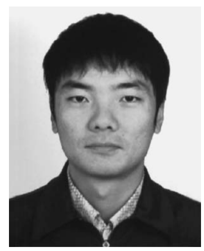

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